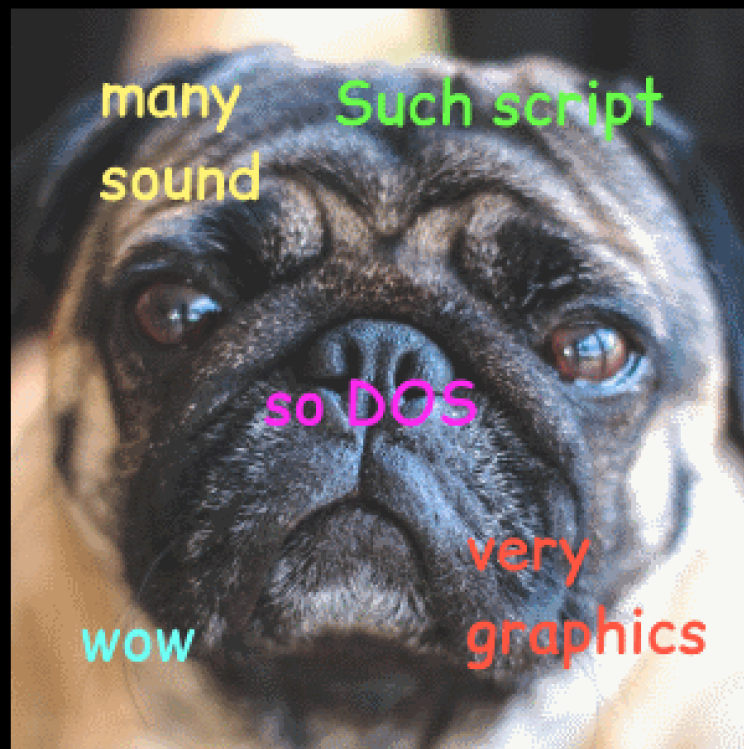


```
D:\>dojs -r EXAMPLES\UCFE\main.js_
```

# DOjs - Javascript for MS-DOS

How a C64 game turned into a Mastodon client for MS-DOS



## Who am I & motivation

---

- \* Who am I & motivation
- \* Creative coding / Processing
- \* IDE / Editor
- \* Features
- \* DOSTodon
- \* Questions

# Who am I

---

Me:

- Goth and metalhead
- Standard nerd
- Gamer
- Retro computer/console collector
- Creative coder (wannabe)
- RL: Embedded software developer
- Pronouns: they/them | he/his



# Motivation

---

## Motivation:

- Lets build a game for the C64
- I know neither 6502 nor HW specifics
- Maybe another platform?
- DJGPP and Allegro are great
- Wait, I need a game idea!
- Ok, build a framework first...
- Oh look: a Javascript interpreter!
- Escalation level > 9000



# Creative coding

---

Inspired by Processing / p5js

[...]  
Processing is an open-source graphical library and integrated development environment (IDE) / playground built for the electronic arts, new media art, and visual design communities with the purpose of teaching non-programmers the fundamentals of computer programming in a visual context.

[...]  
From: [https://en.wikipedia.org/wiki/Processing\\_\(programming\\_language\)](https://en.wikipedia.org/wiki/Processing_(programming_language))

# Creative coding

---

Inspired by Processing / p5js

[...]  
Processing is an open-source graphical library and integrated development environment (IDE) / playground built for the electronic arts, new media art, and visual design communities with the purpose of teaching non-programmers the fundamentals of computer programming in a visual context.

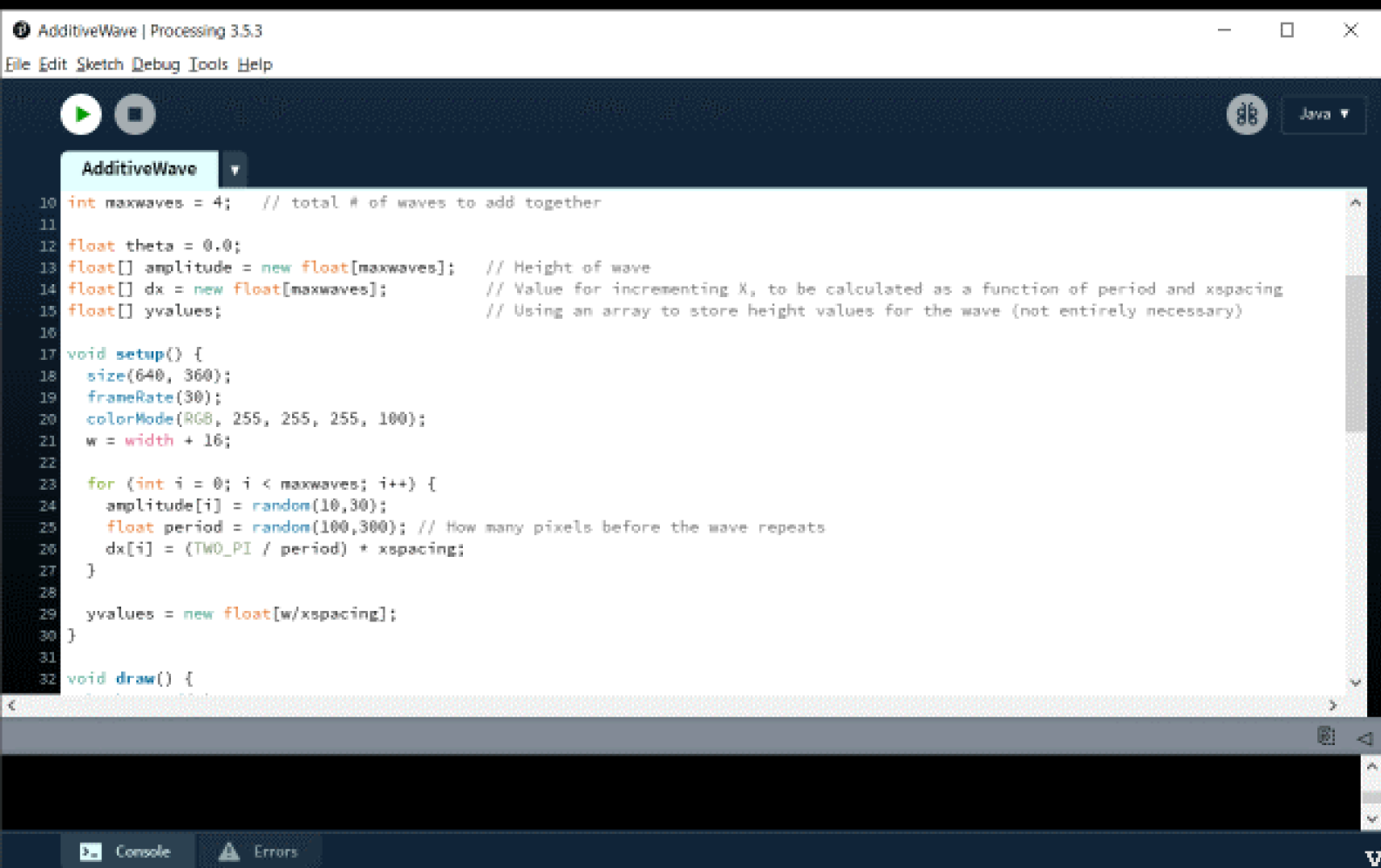
[...]  
From: [https://en.wikipedia.org/wiki/Processing\\_\(programming\\_language\)](https://en.wikipedia.org/wiki/Processing_(programming_language))

**p5.js is a friendly tool for learning to code and make art. It is a free and open-source JavaScript library built by an inclusive, nurturing community. p5.js welcomes artists, designers, beginners, educators, and anyone else!**

From: <http://p5js.org/>

# Processing (screenshot)

## IDE and running sketch



The screenshot shows the Processing IDE interface. The title bar reads "AdditiveWave | Processing 3.5.3". The menu bar includes "File", "Edit", "Sketch", "Debug", "Tools", and "Help". The toolbar contains icons for running (a green play button), stopping (a grey square), and a palette icon. The language dropdown is set to "Java". The sketch name "AdditiveWave" is displayed in a tab. The code editor shows the following code:

```
10 int maxwaves = 4; // total # of waves to add together
11
12 float theta = 0.0;
13 float[] amplitude = new float[maxwaves]; // Height of wave
14 float[] dx = new float[maxwaves]; // Value for incrementing X, to be calculated as a function of period and xspacing
15 float[] yvalues; // Using an array to store height values for the wave (not entirely necessary)
16
17 void setup() {
18   size(640, 360);
19   frameRate(30);
20   colorMode(RGB, 255, 255, 255, 100);
21   w = width + 16;
22
23   for (int i = 0; i < maxwaves; i++) {
24     amplitude[i] = random(10,30);
25     float period = random(100,300); // How many pixels before the wave repeats
26     dx[i] = (TWO_PI / period) * xspacing;
27   }
28
29   yvalues = new float[w/xspacing];
30 }
31
32 void draw() {
```

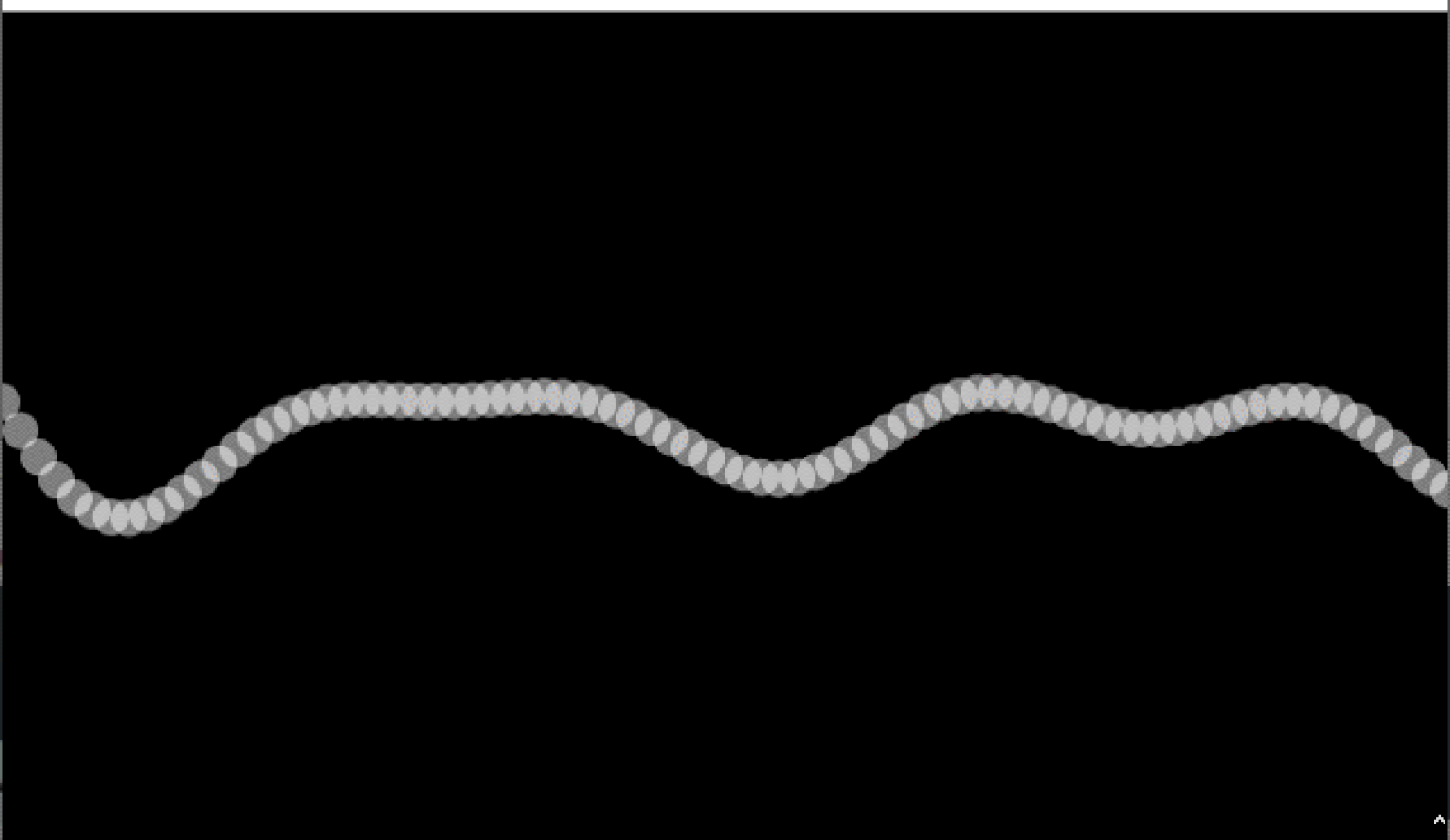
At the bottom, there are tabs for "Console" and "Errors".



# Processing (screenshot)

IDE and running sketch

▶ AdditiveWave



# P5JS example

## Addwave

DOJS 01.2.0 EXAMPLES\ADDWAVE.JS

```

Include('p5');

xspacing = 8;    // How far apart should each horizontal location be spaced
w = 0;          // Width of entire wave
maxwaves = 4;    // total # of waves to add together

theta = 0.0;
amplitude = new Array(maxwaves);    // Height of wave
dx = new Array(maxwaves);           // Value for incrementing X, to be calculated
yvalues = 0;                        // Using an array to store height values

function setup() {
  frameRate(60);
  colorMode(RGB, 255, 255, 255, 100);
  w = width + 16;

  for (var i = 0; i < maxwaves; i++) {
    amplitude[i] = (random(10, 30));
    var period = random(100, 300); // How many pixels before the wave repeats
    dx[i] = ((Math.PI * 2 / period) * xspacing);
  }

  yvalues = new Array(w / xspacing);
}

function draw() {
  background(0);
  calcWave();
  renderWave();
}

function calcWave() {
  // Increment theta (try different values for 'angular velocity' here)
  theta += 0.02;

  // Set all height values to zero
  for (var i = 0; i < yvalues.length; i++) {
    yvalues[i] = 0;
  }

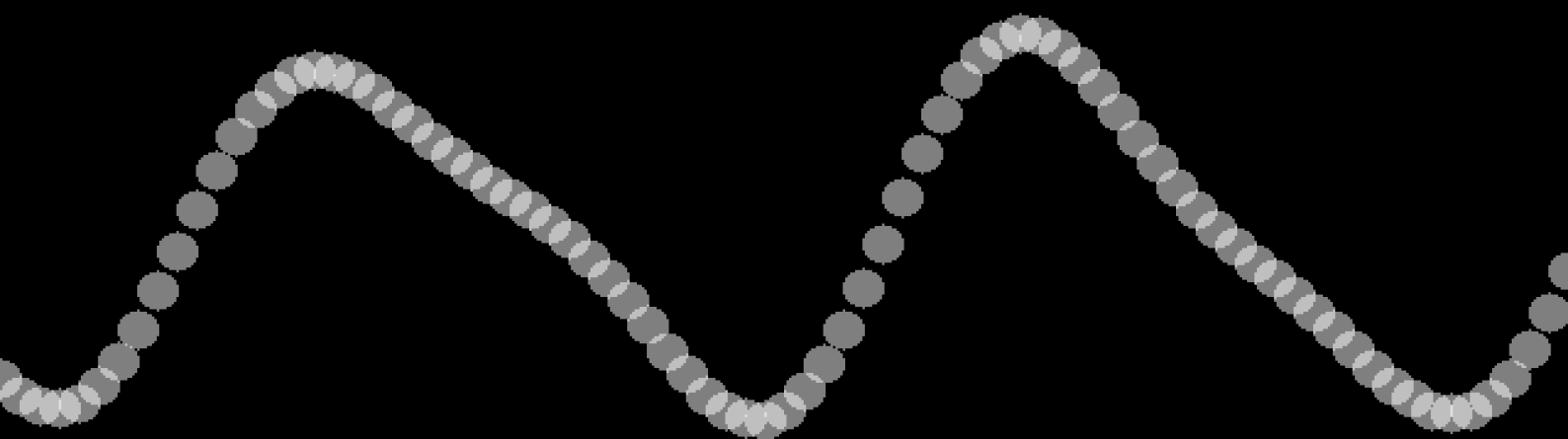
  // Accumulate wave height values
  for (var j = 0; j < maxwaves; j++) {
    var x = theta;
    for (var i = 0; i < yvalues.length; i++) {
      // Every other wave is cosine instead of sine
      if (j % 2 == 0) {
        yvalues[i] += Math.sin(x) * amplitude[j];
      }
    }
  }
}

```

1 Help 2 3 Save 4 Run 5 6 7 Find 8 9 Log V

# P5JS example

Addwave



# Editor

Builtin editor w/ syntax highlight and Copy & Paste

DOJS V1.00    EXAMPLES\ADDWAVE.JS

0034/0040

```
dx = new Array(maxwaves);           // Value for incrementing X, to be calculate
yvalues = 0;                         // Using an array to store height values

function setup() {
  frameRate(60);
  colorMode(RGB, 255, 255, 255, 100);
  w = width + 16;

  for (var i = 0; i < maxwaves; i++) {
    amplitude[i] = (random(10, 30));
    var period = random(100, 300); // How many pixels before the wave repea
    dx[i] = ((Math.PI * 2 / period) * xspacing);
  }

  yvalues = new Array(w / xspacing);
}

function draw() {
  background(0);
  calcWave();
  renderWave();
}
```

1 Help   2   3 Save   4 Run   5   6   7 Find   8   9 Log   10 Exit

# Context Help

Integrated HELP / API doc

DOJS V1.00 EXAMPLES\3DFXCUBE.JS

0139/0005

```

ClearScreen(EGA.BLACK);
for (var i = 0; i < a3dat.arcs.length; i++) {
    var ca = a3dat.arcs[i];
    var ai = CircleArc(a3dat.center, a3dat.center, ca.r, ca.start, ca.start
    if (i % 2) {
        ca.start++;
        if (ca.start > 3600) {
            ca.start = 0;
        }
    } else {
        ca.start--;
        if (ca.start < 0) {
            ca.start = 3600;
        }
    }
}
a3 = new TexInfo(arcer3Bitmap);
// allocate texture memory only once
if (a3TextDest == 0) {
    a3TextDest = FxTexMemGetStartAddress(GR_TMU.TMU0, a3);
}
a3.DownloadMipMap(GR_TMU.TMU0, a3TextDest, GR_MIPMAPLEVELMASK.BOTH);
SetRenderBitmap(null);

```

1Help 2 3Save 4Run 5 6 7Find 8 9Log 10Exit v

# Context Help

Integrated HELP / API doc

DOJS V1.00    EXAMPLES\3DFXCUBE.JS

0139/0005

### ClearScreen(c:Color)

clear the screen with given color.

### Plot(x:number, y:number, c:Color)

draw a point.

### Line(x1:number, y1:number, x2:number, y2:number, c:Color)

draw a line.

### Box(x1:number, y1:number, x2:number, y2:number, c:Color)

draw a box.

### Circle(x1:number, y1:number, r:number, c:Color)

draw a circle.

### Ellipse(xc:number, yc:number, xa:number, ya:number, c:Color)

draw an ellipse.

### CircleArc(x:number, y:number, r:number, start:number, end:number, style:number, c:Color):{"centerX":XXX,"centerY":XXX,"endX":XXX,"endY":XXX,"s

—<UP/DOWN, ESC or DEL>—

1Help   2   3Save   4Run   5   6   7Find   8   9Log   10Exit ^

# Context Help

Integrated HELP / API doc

Documentation

MODULES

3d

3dtx

a3d

color

**gfx**

ipx

joystick

midi

other

p5compat

andin

andout

tcpip

NumberDict

PVector

StringDict

TypedDict

CLASSES

ArcInfo

Bitmap

COMPort

Event

File

Font

FxState

HTTPResult

API Documentation

MODULE

**gfx**

All graphics functions.

[View Source](#)

doc/internal.js, line 208

**Methods**

**INNER**

**Box(x1, y1, x2, y2, c)**

draw a box.

**Parameters:**

| Name | Type   | Description         |
|------|--------|---------------------|
| x1   | number | start x coordinate. |
| y1   | number | start y coordinate. |
| x2   | number | end x coordinate.   |
| y2   | number | end y coordinate.   |

METHODS

Box

Circle

CircleArc

ClearScreen

CustomCircle

CustomCircleArc

CustomEllipse

CustomLine

Ellipse

FilledBox

FilledCircle

FilledEllipse

FilledPolygon

FloodFill

GetPixel

GetScreenMode

Line

Plot

SaveBmpImage

SavePcxImage

SavePngImage

SaveTgaImage

SetRenderBitmap

SizeX

SizeY

TextXY

# Debugging / Logs

## Integrated Logviewer

DOJS V1.00 EXAMPLES\ARCER.JS

0001/0001

```
>>> -----  
>>> DOJS V1.00 (Feb 22 2020 21:13:10) starting with file EXAMPLES\ARCER.J  
S  
>>> Mouse detected  
>>> Sound input: Sound system not installed  
>>> IPX not found.  
>>> DOJS Shutdown...
```

&lt;UP/DOWN, ESC or DEL&gt;

1Help 2 3Save 4Run 5 6 7Find 8 9Log 10Exit

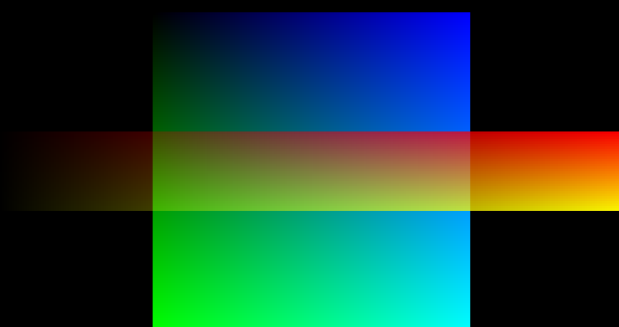




# Colors & transparency

---

- \* 32 Bit RGBA color model
- \* >16mio colors, 255 levels of transparency
- \* HSBA helper function available
- \* full p5js color support (RGB, HSL, HSB)



# Typography

---

- \* GRX font loading
- \* ~390 fonts are included
  - fonts from libgrx
  - fonts from Linux Font Project
- \* for now only horizontal, left to right rendering

The quick brown fox jumps over the lazy dog

The quick brown fox jumps over the lazy dog

The quick brown fox jumps over the lazy dog

The quick brown fox jumps over the lazy dog

**The quick brown fox jumps over the lazy dog**

The quick brown fox jumps over the lazy dog

Της θυγγκ βρωον φοξ φυμπο οωερ της λαζψ δογ

The quick brown fox jumps over the lazy dog

# Files / ZIP archives

---

- \* file system functions
  - read/write files as text or byte wise
  - create/rename/delete files/directories
  - list directories & get file info
- \* ZIP archive support
  - create/extract ZIP files
  - load font, bitmaps, etc from ZIP files
  - load scripts and all support/resource files from single ZIP file
- \* run external programs

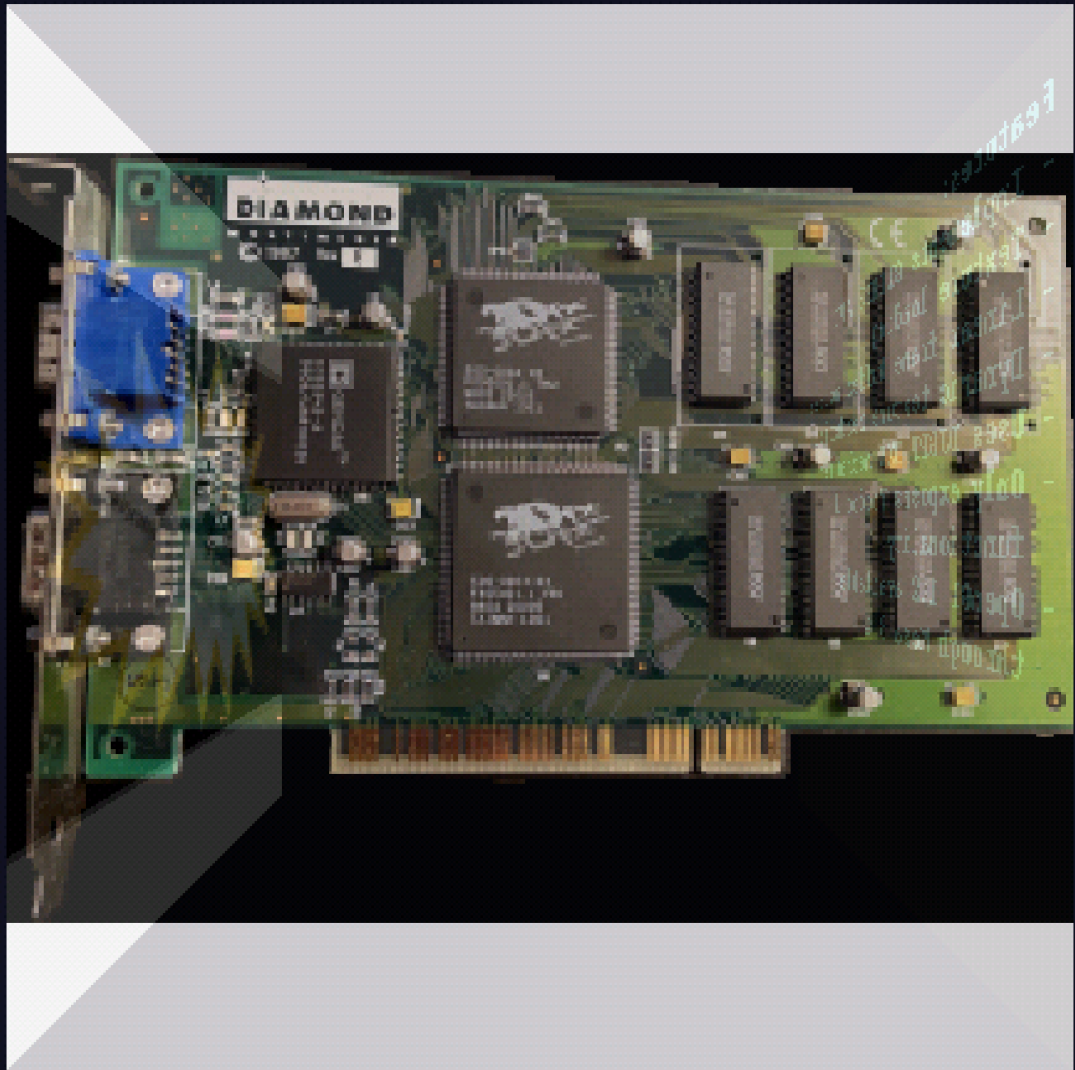
# TCP/IP support

---

Using Watt32

- \* TCP/IP support including:
  - \* TCP client/server sockets
  - \* UDP sockets
  - \* DHCP and DNS
- \* Curl HTTP(S) client
- \* Example sketch with HTTP server

This example polls the ISS position in real time via REST API from <https://www.n2yo.com>





3dfx Interactive was a company headquartered in San Jose, California, founded in 1994, that specialized in the manufacturing of 3D graphics processing units and graphics cards. It was a pioneer in the field from the late 1990s until 2000.

The company's flagship product was the Voodoo Graphics, an add-in card that accelerated 3D graphics.



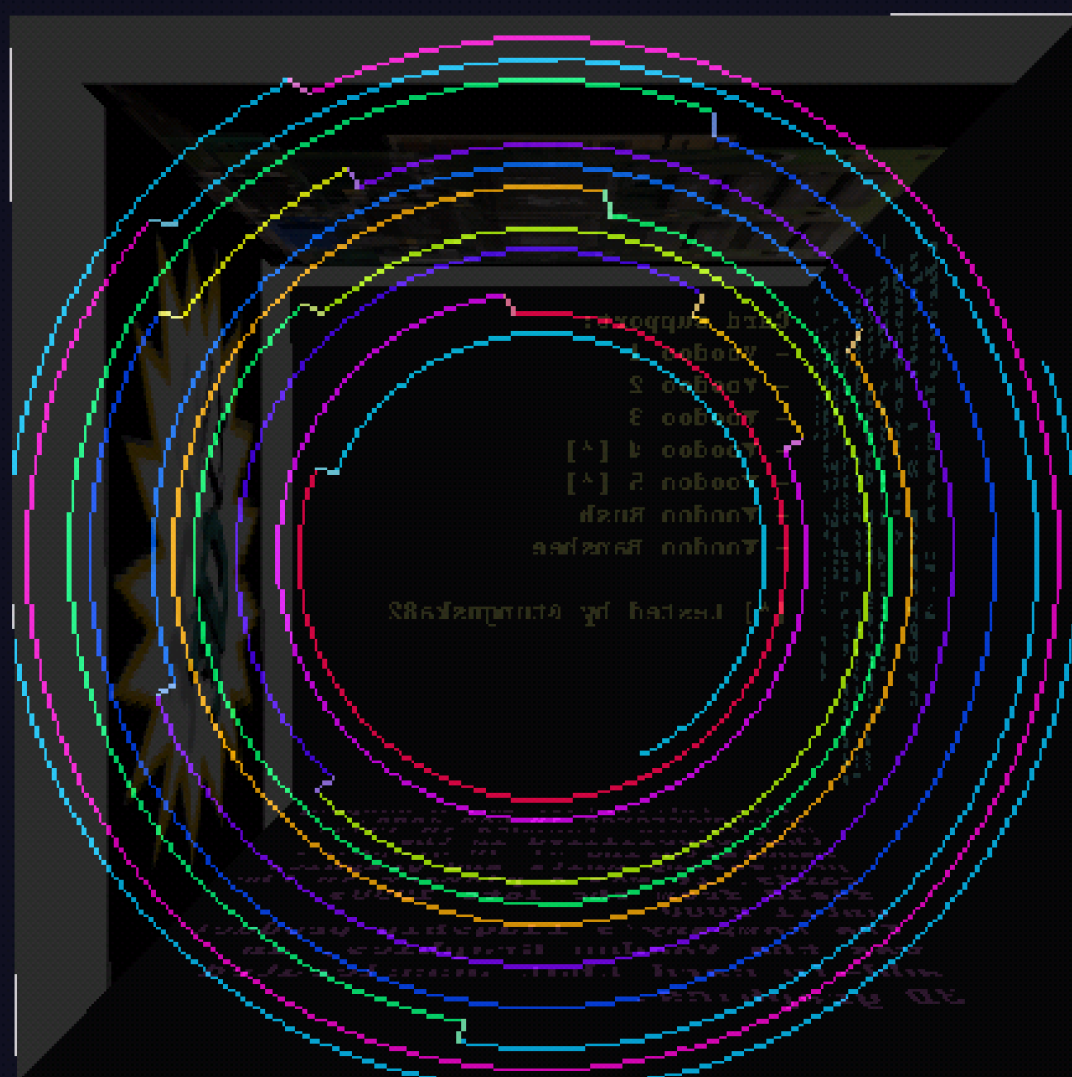
### Features:

- Implements GLIDE3 API
- Texture loading
- Linear frame buffer access
- Dynamic texture rendering
- Uses DJGPP DXE mechanism
- Only exposes Voodoo 1 functionality
- OpenGL 1.2 available through Mesa 3.4

## Card support:

- Voodoo 1
- Voodoo 2
- Voodoo 3
- Voodoo 4 [\*]
- Voodoo 5 [\*]
- Voodoo Rush
- Voodoo Banshee

[\*] tested by @tunguska82



# Features / Infos

---

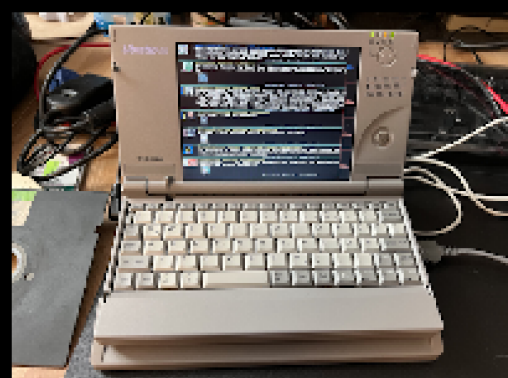
- > Runs on MS-DOS, Win9x, FreeDOS, DOSBox and more
- > Designed for >100MHz Pentium, but starts on 386 as well
- > Original and p5js compatibility API
- > RGB, HSB/HSV colors with alpha channel
- > 2D primitives incl. polygons and arcs
- > Simple text rendering w/ font loading
- > Affine transformations
- > BMP, PCX, TGA, TIFF, JPEG, WEBP, GIF, SVG, JPG2000 and PNG image loading (& saving)
- > Sound & MIDI playback and audio input (WAV, OGG, MOD, S3M, MIDI, SID)
- > Parallel (LPT) and serial (COM) port access
- > 3dfx Voodoo 1/2/3/4/5/Rush/Banshee support with OpenGL and GLIDE3
- > IPX and TCP/IP networking (HTTP(S) support by Curl)
- > Supports remote logging via IPX
- > Mouse, keyboard and joystick support
- > file system access and ZIP file support
- > SQLite3
- > Based on DJGPP, MUJS, Allegro, p5js DZCOMM, WATT32 and Glide3

# DOSTodon

---

- Started as a joke in November 2022
- Supports reading, posting, images, etc.
- Application logic implemented in Javascript
- Runs on MS-DOS, Linux and Win32

-> Short demo (if possible)...

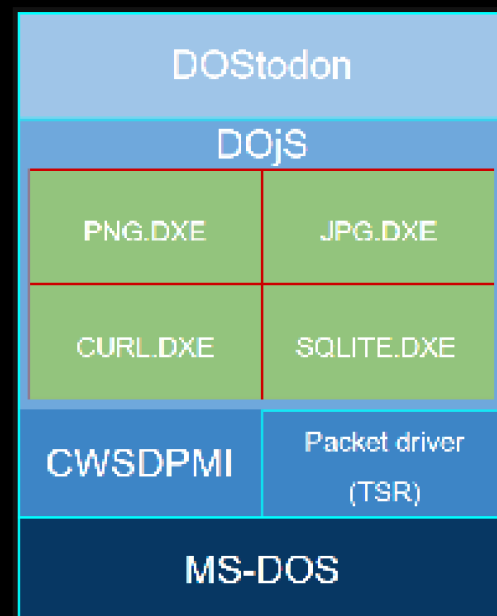


# DOSTodon - Tech Stack

- DOSTodon is implemented in Javascript targeting DOjs
- DOjs is implemented using DJGPP
- It runs on CWSDPMI DOS extender
- TCP/IP is done using builtin Watt32 & packet driver for the NIC

## Plugins:

HTTPS is provided by CURL + mbedTLS  
 Image caching is done using SQLite3  
 WEBP, PNG and JPEG support



# Contact / Website

---

Fediverse/Mastodon:

@dec\_hl@mastodon.social

Github:

<https://github.com/SuperIlu/DOjS>

<https://github.com/SuperIlu/DOSTodon>